

Automatic standard building category classification from smart meter data – A supervised learning approach

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ABSTRACT

Increased availability of smart meter data offers better insight into buildings' electricity usage. By classifying smart meter data by building type and presence of heating appliances, we can efficiently gain metadata about the buildings that is useful for research, grid planning, and energy efficiency policy employment. However, current smart meter classification approaches are largely based on limited datasets and building classes, or on unsupervised methods that don't align with standard building categories and offer limited control over grouping. This article presents a supervised automatic building category classification approach for labelling smart meter data from buildings into standard building categories in the Norwegian building regulations (TEK17), and whether they have electric heating or not. 82 novel physics-based domain features are presented which can be extracted from any hourly electricity smart meter data series from buildings with a duration of months-years. The features are specifically designed to identify the building and heating type of a smart meter data series by capturing patterns such as seasonality, daily usage trends, similarities with standardized building load profiles, temperature dependency, and other domain-specific characteristics. The classification approach is trained and tested on a large dataset of 2724 buildings from 12 different building categories, both residential and non-residential, and correctly identifies the heating type and building category of unseen Norwegian smart meter data from buildings in 84 % of the test cases. The approach is generalizable to meter data from other Norwegian buildings and is also tested on buildings from other climate zones. The proposed method for smart meter data classification is proven to have high accuracy and applicability for extracting metadata for both residential and non-residential buildings in Norway.

1. Introduction

In 2021, the operation of buildings was responsible for 30 % of final global energy consumption and 27 % of total energy sector emissions (out of which 8 % is related to direct emissions from buildings, while 19 % refers to emissions from generation of heat and electricity consumed by buildings) [1]. The electrification of buildings has been identified as a key alternative to achieve a more sustainable energy system and mitigate the corresponding emission of gases that result in climate change [2]. To tackle these challenges, more knowledge about building energy use is needed. Due to recent roll-out of smart grid infrastructures and advanced digital metering systems, building energy data has become more available. Norway mandated the installation of smart electricity meters for all electricity consumers as part of advanced metering systems (AMS) by 2020 [3]. These meters record customers' hourly

electricity usage and transmit data to grid companies, and can, if analysed, provide more information about electricity use patterns for different electricity consumers. The use of electrical heating appliances is widespread in Norway due to historically low electricity prices [4]. It is estimated that electricity for heating makes up more than 60 % of the total electricity consumption in Norwegian buildings [5]. The use of electricity for heating hence contributes to high peaks in the electricity grid during the colder winter days [6]. Buildings with electrical heating hence typically have a stronger strain on the electricity grid compared to buildings with non-electric heating. To address the challenges of increased electricity demand in the grid, and to utilize the availability of more electricity data from buildings, the implementation of classification methods for electricity measurements becomes valuable. By classifying smart meter data by building type, we can efficiently gain metadata about the buildings that is useful for research, grid planning,

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and energy efficiency. Unsupervised methods have been applied in various ways to classify smart meter data, as demonstrated in [7] and [8], where clustering techniques were used to group households based on their demand response potential, and in [9] and [10] which looked at grouping of buildings into distinct consumer classes to design tailored tariffs aimed at reducing grid strain. Other examples include [11] which investigated K-means clustering applied to clustering of 14 non-residential buildings on a university campus (of types office/library/teaching) and clustered into two different consumer groups. The effectiveness of unsupervised methods/clustering relies on multiple factors and can be measured by diverse indicators to assess how well the clusters represents the data, including the silhouette score [12], Davis Bouldin index and the Dunn index [13]. While these indicators can tell us how well the buildings within each group fit together, unsupervised clustering techniques may group together buildings have several similarities but serve different purposes. Another challenge with unsupervised methods is that the number of optimal clusters is generally not known. There are numerous methods for estimating the optimal number of clusters but depending on the problem, unsupervised methods may not be the most suitable for assigning a class to a building load profile. The classification of buildings' smart meter data has historically been considered an unsupervised machine learning task [14,15] due to both the lack of properly labelled smart meter data and the challenge of determining the correct or optimal number of clusters. However, unsupervised methods are not necessarily the best approach when aiming to classify buildings into set, pre-defined building categories. In Norway standardized building category groups are used in the Norwegian building code TEK-17 [16], and in the standard for energy calculations of buildings in Norway [17] and divide buildings/part of buildings into 13 main categories ("small house", "apartment", "school", "office", "nursing home", etc.). Each category has specific energy efficiency requirements, making it essential to understand the composition of these categories within an area for effective grid and energy planning. Classifying buildings' smart meter data into these standardized categories can provide a more accurate and practical categorization of buildings. For instance, while schools and kindergartens may exhibit similar energy consumption patterns, a notable distinction arises in the heating systems employed in these building categories. Kindergartens often have point source heating, while schools often utilize water-borne heating systems, providing them with greater flexibility to alter heating sources and control strategies. Likewise, a nursing home might have similar energy consumption profiles as an apartment building, yet an important difference between these types of buildings are the ownership structures. Publicly owned buildings may be in a different economical position compared to privately owned residential properties, and policymakers wield more influence in enforcing legislation and refurbishments in the former. Cabins equipped with electrical heating may have similar technical systems compared with single family houses, but their geographical location introduces unique considerations. Situated in areas with limited grid capacity and sparse occupancy on most days, areas with many cabins may experience a pronounced coincidence factor during holidays when numerous consumers are present. These differences may not be captured when using unsupervised clustering algorithms, and the clusters generated with unsupervised clustering algorithms may be capturing unintended attributes about building electric load patterns than anticipated. Supervised classification methods provide an alternative to clustering that offers more control of the assignment of labels to building load profiles. While supervised classification is widely used in many classification tasks, only a few supervised classification of electricity load profiles from buildings have been demonstrated in research for deriving the building class and meta data about buildings, likely since supervised learning techniques require a large amount of labelled data for training, validation, and testing with clear segmentation. One example where supervised classification has been used on smart meter data is for prediction building energy consumption, such as in [18] for an office building in New York to categorize energy

consumption into one of four predefined patterns, representing weekend use, night use, and weekday use during summer/winter. Supervised classification has also been used to classify the energy performance of 2717 commercial buildings in Hong Kong from meta data and measured energy use when their size were unknown, achieving an accuracy of 78.8 % with random forest. This approach did however rely on meta data as input features which are usually not available within smart meter databases or in publicly available statistics elsewhere. Similarly, [19] demonstrated a classification model based on artificial neural network for classifying energy performance certificated of Italian buildings. The input features of this classification were mostly meta data present in the buildings' EPCs, like U-values, area, degree days and volume etc, and the only measured input value was CO₂-emissions from the buildings. The model showed to receive up to 99 % accuracy depending on included features but could only achieve an accuracy of 75 % without measured/calculated data of energy use or CO₂-use, which is directly linked to the EPC of the building.

There are a few examples where meta data from buildings has been extracted from smart meter data using supervised classification. One supervised classification of building categories used a semi-supervised approach to group 3 months data from 114 non-residential buildings into 17 different building typologies using a follow-the-leader clustering approach and achieved an accuracy of 80 % [20]. In another example, a supervised learning approach with domain-informed and domain-agnostic features were used to estimate the country of buildings from 4 different countries [21]. Supervised learning was used in two studies with several thousand smart meters from Ireland to extract metadata about the households such as heating type, floor area, age and number of inhabitants. 48 (number of bedrooms, 4 classes) – 84 % (single/not single occupant) accuracy dependent-independent data classification (DID) and probabilistic regression [22], and in [23] to estimate the household size class (number of adults and children) of the Irish dwellings. The latter did however show that that the smart meter data alone was limited to distinguish the household category and achieved a low accuracy. To the best of the authors knowledge, there exist no method or application for classifying the standard building category and heating type of buildings from their smart meter data, and there is limited research on supervised classification of smart meter data for extraction of building meta data in datasets containing both residential and non-residential buildings.

1.1. Contributions

The primary research problem addressed in this article is how to collect metadata for buildings' smart meter data when it is not readily available, specifically their building category and heating type. Collecting information about buildings is a difficult and time-consuming task that typically requires manual effort. To address this, there is a need for a method that is generalizable to smart meter data of different durations, without relying on additional metadata, that can be applied to buildings of varying locations, types, sizes, ages, and energy performance levels."

This article introduces a novel supervised classification method for segmenting smart meter data from buildings into standardized categories. While the true number of classes is considered unknown in most classification approaches of smart meter data, this approach uses the standard building categories from the Norwegian building regulations as labels to develop a model that can differentiate between different building categories, including both residential and non-residential buildings. In short, the contributions of this paper are:

- A supervised classification approach for segmenting smart meter data from buildings into standardized building categories, including both residential and non-residential buildings as well as their heating class (electric and non-electric).

- The classification approach is trained and tested on a large and novel dataset of hourly data from more than 2700 Norwegian buildings from 11 building categories (including 4 residential categories and 7 non-residential categories). To the best of the authors knowledge, there are no other building category classification approaches in research based on a dataset on this scale, and most approaches found in the literature are focused on either residential or non-residential buildings separately.
- For the classification, 82 physics-based domain features are extracted from each of the smart meter time series from the buildings. The features can be extracted from any hourly electricity smart meter data series from buildings with a duration of months-years.
- A comparison between the proposed supervised method and an unsupervised classification of the smart meter data using K-means clustering is presented, showing how a supervised approach is more suited for classifying buildings into their standardized building categories.
- The method is generalizable to other Norwegian buildings outside the dataset and is in addition tested against external, openly available datasets of electricity consumption in buildings from other locations and climate zones and compared the model results from the suggested approach against unsupervised methods for building category classification.

2. Methodology

This article presents a method to classify buildings' AMS-data into a building category and heating type (electric/non-electric). This section describes the dataset used to develop and test this method, the labels of the buildings in the dataset, the physics-based domain features extracted for each building, and different models tested for the classification problem.

2.1. Dataset and labels

To develop and test the classification method, a large data set containing files with several years of hourly electricity and outdoor temperature measurements from 2724 buildings located in Norway is used. The dataset consists of both openly available data and classified data provided by Drammen Municipality, FME ZEB [24], FME ZEN [25], Elvia, Statkraft, Risvøllan Housing Association, Sikom, Statsbygg, and other project partners. More information about the file structure of the building files is described in [26]. Every building in the data set is labelled with a *building category* – describing the primary use of the building – which are the standardized building categories used in the Norwegian building code TEK-17 [16], and in the standard for energy calculations of buildings in Norway [17]. Each building is also labelled with a *heating type* (describing whether the building has electric/non-electric space heating) and with a resulting *building type* (combination of the *building category* and *heating type*). A summary of the labels and the support within each building type is given in Table 1.

2.2. Feature generation

Smart meter data collected from different sources will often have varying start dates, end dates, seasons, and durations. These discrepancies in time series data can lead to inaccurate predictions, as it may result in comparing apples to oranges. One can consider a scenario where image recognition is used to differentiate between schools and apartment blocks based on their smart meter data. If all school buildings, except one, have data spanning four years, while all apartment blocks have data covering only one year, the single school with only one year of data might be incorrectly classified due to the difference in data duration. To address this issue, feature extraction can be employed. Feature extraction involves identifying and extracting relevant patterns and characteristics from the smart meter data, which can then be used to

Table 1

Building type labels and the number of buildings in the dataset.

	Building category in TEK17	Building category label	Heating type (Electric Heating (EH)/ non-electric heating (NEH))	Building type label	Number of buildings
Labels for building category and type (not present in dataset in grey)	House	Hou	NEH	Hou_NEH	67
			EH	Hou_EH	473
	Apartment	Apt	NEH	Apt_NEH	1344
			EH	Apt_EH	400
	Apartment blocks*	Apb	NEH	Apb_EH	53
			EH	Apb_NEH	51
	Cabin/Holiday house	Cab	NEH	Cab_NEH	0
			EH	Cab_EH	128
	Office	Off	NEH	Off_NEH	54
			EH	Off_EH	7
	Nursing home	Nsh	NEH	Nsh_NEH	21
			EH	Nsh_EH	6
	Hotel	Htl	NEH	Htl_NEH	7
			EH	Htl_EH	0
	Kindergarten	Kdg	NEH	Kdg_NEH	11
			EH	Kdg_EH	20
	School	Sch	NEH	Sch_NEH	32
			EH	Sch_EH	16
	Shop	Shp	NEH	Shp_NEH	28
			EH	Shp_EH	0
	Hospital	Hsp	NEH	Hsp_NEH	6
			EH	Hsp_EH	0
	Culture	Cul	NEH	Cul_NEH	0
			EH	Cul_EH	0
	Sport	Spo	NEH	Spo_NEH	0
			EH	Spo_EH	0
	University	Uni	NEH	Uni_NEH	0
			EH	Uni_EH	0
				Total	2724

*In the database and in the Norwegian building codes TEK17, apartments (section within an apartment block) and apartment blocks are both labelled the same. In this dataset, these are separated into apartments and apartment blocks.

make the building time series data comparable. By focusing on key features such as average energy consumption, seasonal variations, and usage patterns, we can reduce the bias in the dataset caused by differences in e.g. durations and improve the accuracy of predictions. This approach ensures that the classification model compares similar features across different datasets and reduce the data quantity and the running time. Here, 82 physics-based domain features are generated for each building from its electricity load and temperature time series. The features are categorized into 5 feature groups:

- Non-normalized electricity load features.
- Electricity load variation features.
- Seasonal differences features.
- Average daily profile features.
- Standard load profile correlation features.

A list of all features within each group is given in the following sub chapters.

2.2.1. Non-normalized electricity load features

Four features describe the non-normalized data of the buildings' electricity load. These features capture the size of the peak load, and mean electricity use and the relationship between the electricity load and outdoor temperature. These features extract information about the size of the building (which is important to gain information about the building type) and the temperature dependency of the load. The last two features look at the relationship between the electricity use and the

outdoor temperature. The first is the outdoor temperature at the peak electricity load, and the second is the imported electricity at the minimum outdoor temperature. A summary of the Non-normalized electricity load features are given in Table 2.

2.2.2. Electricity load variation features

The 32 electricity load variation features are normalized electricity load features that look at peak load (max), average load, the relationship between them, and the hour of the day when peak load occurs for different subsets of the data – for example, during the wintertime, summertime, or for the entire dataset. The subsets are chosen to examine season and temperature dependent variations. The winter season is defined as December-February, spring as March-May, summer as June-August and autumn from September-November. The electricity load variation features are summarized in Table 3.

2.2.3. Seasonal differences features

Seasonal variation and temperature dependency can provide an indication of whether the building uses electricity for space/ventilation heating. To further capture such effects, 6 seasonal differences features are added. They are intended to complement the electricity load variation features and to specifically investigate the relationship between the summer and winter features as these are the most important features to evaluate whether a building has electric heating or not. These features examine the differences in the mean and max in summer and winter. The seasonal differences features are summarized in Table 4.

2.2.4. Average daily profile features (normalized data)

The 16 “Average daily profile features” are features which are generated to extract traits from the daily load profile are given in Table 5. These features represent specific patterns that explain user behaviour and variations throughout a day that can indicate the type of building from its electricity load profile. The features are selected to capture the most important elements that can indicate the building category based on domain knowledge. These features provide a stronger insight compared to just giving the daily load profile values (Fig. 1).

2.2.5. Standard load profile correlation features (normalized data)

To capture the similarities between the daily user patterns in buildings and the user patterns of similar building types, 24 standard load profile correlation features are extracted as shown in Table 6. The standard load profile correlation features describe the extent of correlation of daily load profile of each building with standard load profiles of different building types. The standard load profiles are generated using the PROFet model [25]. PROFet is an aggregated load profile generator which can predict hourly load profiles for both thermal loads and electric loads, based solely on outdoor temperatures and building area. PROFet is based on panel regression of energy measurements of buildings from trEASURE, a database, of monitored buildings, mostly connected to district heating. PROFet can estimate the typical load profile of an area based solely on building area input (for 12 building categories and 3 energy efficiency levels as described in the categorization) and outdoor temperatures [25,27,28]. PROFet has been used to create standard load profiles for 11 building categories (all building categories

Table 2
Non-normalized electricity load features.

Feature notation	Description
Ellmp_actual_max	The actual (non-normalized) peak load of the imported electricity load profile of the building
Ellmp_actual_mean	The actual (non-normalized) mean value of the imported electricity load profile of the building
Tout_at_Ellmp_max	The outdoor temperature when the electricity peak load occurs.
Ellmp_at_Tout_min	The electricity load when the minimum outdoor temperature occurs.

Table 3
Electricity load variation features.

Feature notation	Description	Subset of data
Ellmp_max	The actual peak load of the imported electricity load profile of the building	All data, winter, spring, summer, autumn, Weekdays, Weekends, the outdoor temperature is below 10C
Ellmp_mean	The mean value of the imported electricity load profile of the building	
Ellmp_max_vs_mean	The maximum peak load divided by the mean of the peak load profile of the building.	
Ellmp_max_hour	The hour of day when the peak load occurs.	

Table 4
Seasonal differences features.

Feature notation	Description
Ellmp_max_vs_summer_max	The maximum peak load divided by the peak load during the summer
Ellmp_mean_vs_summer_mean	The mean electricity load divided by the mean peak load during the summer
Ellmp_winter_mean_vs_summer_mean	The mean electricity load during winter divided by the mean electricity load during summer
Ellmp_max_vs_winter_max	The peak load divided by the peak load during the winter
Ellmp_mean_vs_winter_mean	The mean electricity load divided by the mean electricity load during the winter
Ellmp_winter_max_vs_summer_max	The peak load during the winter divided by the peak load during the summer.

Table 5
Average daily profile features.

Feature notation	Description
daily_average_profile_max	A. Maximum value during the average day.
daily_average_profile_max_hour	B. Hour when maximum value (A) occurs
daily_average_profile_min	C. Minimum value during the average day.
daily_average_profile_min_hour	D. Hour when minimum value (C) occurs
daily_average_max_over_min	Maximum divided by minimum value (A/C)
daily_average_median	E. Median value of the daily load profile.
daily_average_profile_max_after	The value at the hour after the maximum value (value at B + 1)
daily_average_profile_max_before	The value at the hour before the minimum value (value at B-1)
daily_average_std	Standard deviation of the daily average profile
daily_average_var	Variation of the standard profile
daily_average_profile_second_max	F. The second highest hourly value during the average day.
daily_average_profile_second_max_hour	G. The hour when the second highest value occurs.
daily_average_profile_max_change	F. The hour when the change is biggest (positive) from the past hour to the current hour.
daily_average_profile_max_change_hour	G. The hour after the biggest (positive) change during the day.
daily_average_profile_min_change	H. The highest negative change during the average day from the past hour to the current hour.
daily_average_profile_min_change_hour	I. The hour after the largest negative occurs.

in Table 1, excluding cabins, industry and cultural buildings) using a standard weather profile from Norwegian Standards NS3031 [29]. To create standard profiles that represent the average energy efficiency standard of the building stock, it is assumed that the buildings are 82 %

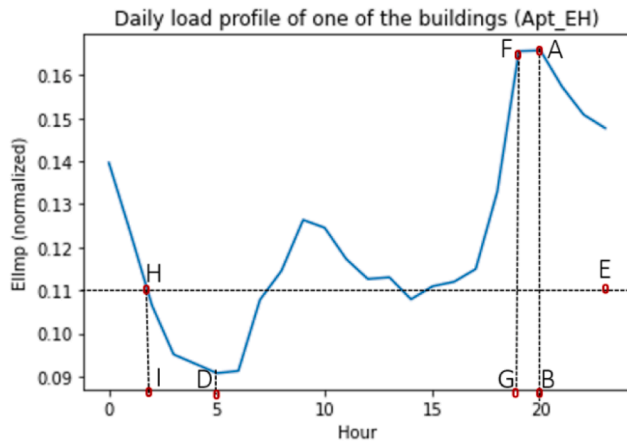


Fig. 1. The average daily load profile of a building and indication of where the features in Table 5 are collected from the daily profile.

Table 6
Standard load profile correlation features.

Feature notation	Description
correlation_daily_SLP_*_EH	Correlation between the daily load profile of the building and the standard category with electric heating
correlation_daily_SLP_*_NEH	Correlation between the daily load profile of the building and the standard category without electric heating
*One of the following category abbreviations (see Table 1): Apt, Hsp, Hou, Htl, Kdg, Nsh, Off, Oth, Sch, Shp, Spo, Uni	

“regular” and 18 % “efficient”. To make standard load profiles for buildings with and without electric heating, it is assumed that buildings with electric heating has an electricity consumption equal to the total demand of the buildings (including electric specific demand, domestic hot water heating and space heating). It is also assumed that buildings without electric heating have an electricity consumption equal to the energy demand for electric specific loads. An example of the daily standard load profiles for apartments and schools with and without electric heating is shown in Fig. 2.

The standard load profiles are generated for one year before they are normalized. Average hourly daily load profiles are then created. The average daily load profiles for some of the standard building categories are shown below. Several patterns can be observed from these figures. Buildings with NEH tend to have higher normalized loads during the average day due to less variations during the year compared to buildings with electric heating, where there are large variations in the peaks due

to differences in the outdoor temperatures throughout the year. Residential buildings typically have a small peak in the morning and a higher peak in the evening, while service buildings have a peak during the day. The rise in energy consumption in the mornings typically starts earlier in buildings with electric heating compared to buildings without electric heating due to the time needed to heat buildings in the morning, and due to the electricity need for hot water in residential buildings. Standard load profile features are calculated as the correlation between each of the standard daily load profiles and the average daily load profile of the building. There is one feature for the correlation between the building and each of the available load profiles.

2.3. Classification models

For the classification task of assigning building category and building type labels to previously unseen electricity time series data from buildings, a selection of machine learning classifiers were explored utilizing the Scikit Learn library in Python version 1.3.0 [30]. The tested classifiers include the following with default parameters: Decision Tree Classifier [31], RandomForest Classifier [32], Support Vector Machines Classifier [33], Gradient Boosting Classifier [34], AdaBoost Classifier [35] and CatBoost Classifier [36]. In addition, the best performing from this list is evaluated against Unsupervised K-means clustering [37] and Soft-voting ensemble learning [38].

2.4. Training and testing

To train and test the classification methods, the dataset comprising 2724 buildings has been randomly split with an 80/20 partition into the training set (2179 buildings) and a test set (545 buildings).

2.5. Performance metrics

To evaluate the performance of the different classification models, a diverse set of performance metrics are employed. These metrics serve as quantitative measures to gauge various aspects of model performance, including accuracy, accuracy when providing two possible labels, precision, recall and F1-score. By applying these performance metrics, it is possible to quantitatively get an understanding how well the models can correctly label previously unseen data. The performance metrics are described in Table 7.

3. Results

3.1. Prediction of building type

Different models are applied on the dataset with extracted features to classify the *building type* (building category + heating type). Classification of the test set has been performed based on training the models

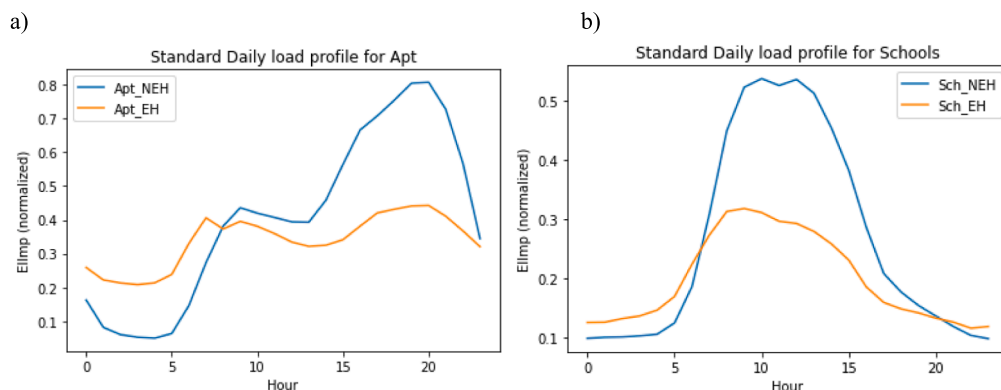


Fig. 2. Standard daily load profile of different building categories (a) with and (b) without electric heating.

Table 7

Definition and description of performance measures.

Metric	Equation	Description
Accuracy (Acc) (32)	$\frac{TP + TN}{TP + TN + FP + FN}$	The share of predictions that were correct
Top-2 accuracy	$\frac{A_{1-2}}{TP + TN + FP + FN}$	The share of predictions that were correct in one of the two most likely classes.
Precision(33)	$\frac{TP}{TP + FP}$	The share of predicted positive that were correct.
Recall (33)	$\frac{TP}{TP + FN}$	The share predictions that should have been predicted as true that were correct.
F1-score (33)	$2 \cdot \frac{prec. \cdot recall}{prec. + recall}$	Combination of precision and recall that strongly penalizes low precision or recall.
Example with classification of a building as “Office”.	Predicted Office	Not Office
Actual	Office	True
	Not Office	False
	Office	Positive
	Not Office	Negative
	Office	(TP)
	Not Office	(FN)
	Office	False
	Not Office	True
	Office	Positive
	Not Office	Negative
	Office	(FP)
	Not Office	(TN)

described in section II. C. on the training set. The results of the performance in correctly identifying the labels of the unseen training set are described in Table 8 and in the confusion matrices of the test set as shown in Fig. 3. The results show that the Random Forest Classifier and CatBoost Classifier perform the best overall on all performance metrics. They achieve similar results, with an overall accuracy of approximately 0.84. When considering two possible labels (top 2 accuracy), these models achieve an accuracy of 0.92, meaning there is a 92 % chance that one out of the two provided categories is the correct label for the given load profile. They outperform each other on different metrics, but the differences are minimal. Given this information, the Random Forest Classifier may be preferred over the CatBoost Classifier as it is a much faster algorithm, while still offering similar overall performance. The Gradient Boosting Classifier and Decision Tree Classifier also provide satisfactory results with accuracies of 0.77 and 0.81, respectively, while Support Vector Machines (SVM) classifier and AdaBoost Classifier show poor performance, with accuracies of 0.60 and 0.28 respectively. Examination of the confusion matrices reveals a tendency for these classifiers to disproportionately allocate unseen data labels to classes abundant in the training set, indicating a bias towards unbalanced data.

Analysis of the confusion matrices for the Random Forest Classifier and CatBoost Classifier indicates minimal confusion between residential and commercial buildings overall. However, within these categories, there exists confusion among specific subcategories.

3.2. Prediction of building category

In another experiment, the best performing models, Random Forest Classifier and CatBoost Classifier, are tested on how well they perform in predicting only the *building category* without the heating type (electric/non-electric). The results are summarized in Table 9 and the corresponding confusion matrix in Fig. 5. The overall performance on

Table 8

Performance of different models on the test set when predicting building type.

Model	Acc.	Acc. top 2	Precision	Recall	F1	Run time
Decision Tree	0.765	0.771	0.782	0.765	0.767	0.4 s
Random Forest	0.844	0.925	0.827	0.844	0.832	2.8 s
Gradient Boosting	0.809	0.897	0.805	0.809	0.801	179.4 s
SVM	0.596	0.000	0.453	0.596	0.496	0.2 s
Ada Boost	0.281	0.325	0.492	0.281	0.180	1.9 s
Cat Boost	0.842	0.919	0.835	0.842	0.834	121.9 s

estimation of the building category is approximately 0.89 in accuracy, precision, recall and F1-score for both models, while the top-2 accuracy is 0.97–0.98. The performance is hence better for the classification approach when only predicting building category and not also the heating type. This is likely due to a combination of less complexity as well as uncertainty in the labelling of the heating categories of the buildings.

3.3. Prediction of building type with feature selection

To check if there are more optimal solutions with a different combination of features, Forward Sequential Feature Selection (FSFS) is applied to find best performance from various subsets of 82 features. In this technique, an iterative model is built by adding one feature at a time, starting from an empty set of features. At each iteration, the algorithm evaluates the performance of the model with the addition of each feature and selects the one that results in the best performance according to accuracy. This process continues until all features are included. Fig. 6 shows the evolution of the best accuracy on the test set when the number of features included is varied from 1 to 82. The performance appears to reach an elbow when 7–12 features are included. Table 10 shows the top 9 features that are included stepwise and the corresponding improvement in performance.

3.4. Comparison with ensemble learning

Ensemble models in classification problems involve combining multiple individual classifiers to generate a more accurate and robust prediction than any single model could achieve alone. Common ensemble techniques include bagging, boosting, and stacking. Ensemble models are widely utilized in various domains due to their ability to improve classification accuracy and generalization while reducing overfitting. To estimate if the classification accuracy can be improved through ensemble learning, an ensemble voting classifier with soft voting is investigated. Two ensemble models are tested: one which combines the Random Forest and CatBoost classifiers, and one where also Gradient Boosting is included. Ensemble models can be affected by class imbalance in data. To investigate if this is the case for this classification problem, a second test is performed where the ensemble models are applied on a subset of the dataset with only residential buildings. The results of applying ensemble model on the entire dataset and only on residential buildings in the dataset is shown Table 11. The results show that the ensemble model has a performance lower but close to the performance of using the Random Forest or CatBoost Classifiers alone (approximately 84 % on the entire dataset as shown in Table 11 for both the entire dataset and for the subset containing only residential buildings. A reason for this may be that these models associate a very high value or importance to the same set of features as shown in Fig. 4, and combining them does not provide any advantage in improving the performance measures (Table 12).

3.4.1. Comparison with k-means clustering

K-means clustering is a popular unsupervised learning algorithm used to partition a dataset into clusters based on similarity. It works by iteratively assigning data points to the nearest cluster centroid and then recalculating the centroids based on the mean of the points assigned to each cluster. This process continues until the centroids converge or a specified number of iterations is reached. In classification, the clusters generated by k-means can represent different classes or categories within the data. The elbow method and silhouette score are both techniques used to help find the optimal number of clusters in k-means clustering. The elbow method helps find the best number of clusters in k-means by looking for a bend (elbow) in the plot of cluster number vs. within-cluster sum of squares. The silhouette score measures how close each point in one cluster is to points in the neighbouring clusters, indicating cluster quality. Higher silhouette scores and noticeable bends

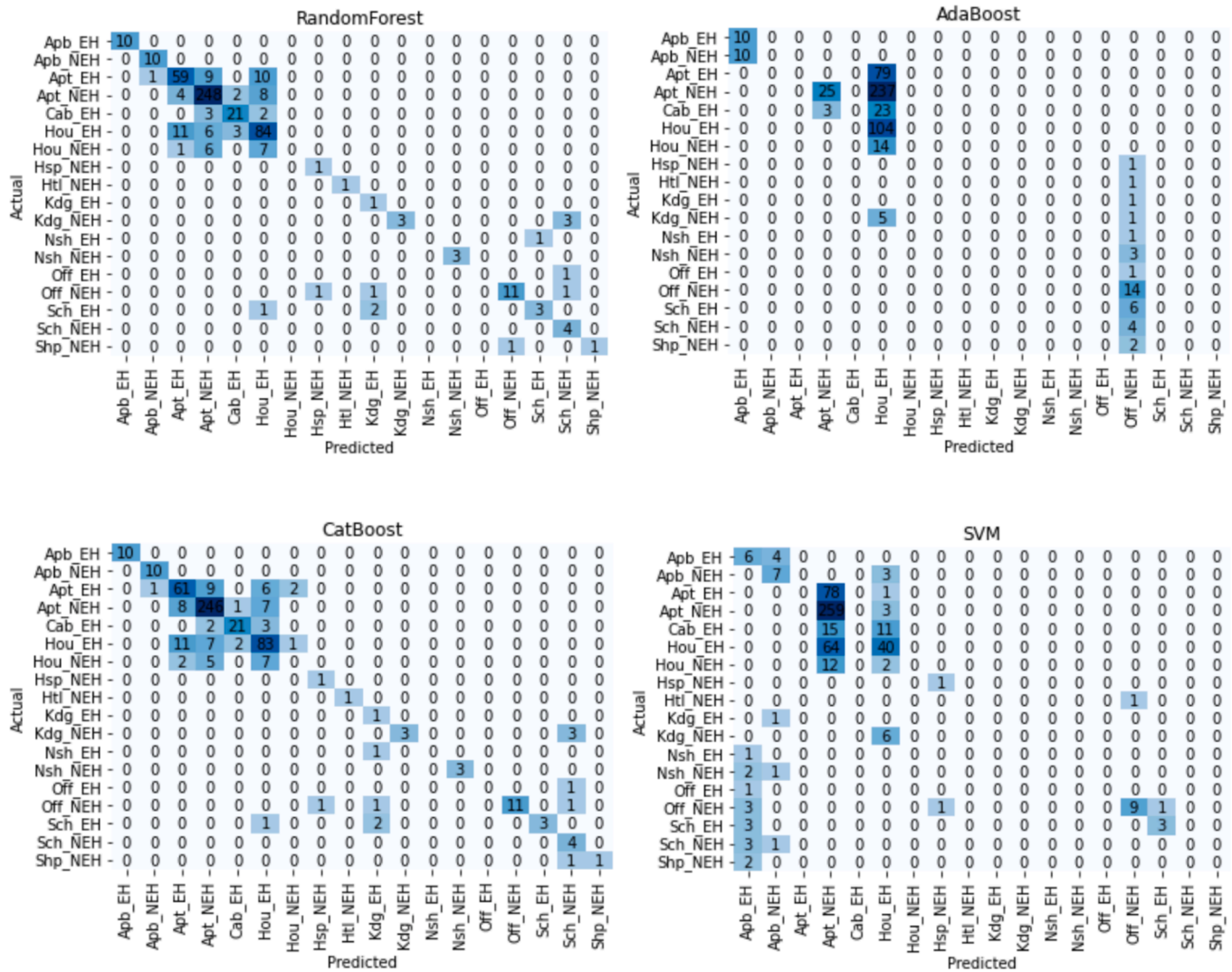


Fig. 3. Confusion matrix of building type classification with different classifiers. Fig. 4 shows feature importance exhibited by the CatBoost, Random Forest and gradient boost classifiers.

in the elbow plot suggest better clustering solutions.

The elbow method and silhouette scores were calculated for different numbers of k on the dataset with results shown in Fig. 7. The Elbow method suggest the number of optimal clusters for the given dataset is around 6. Fig. 8a shows how the buildings in the test set are assigned to different clusters when 6 clusters are used. It shows that residential buildings are all grouped together, while commercial buildings are grouped in three other groups. Fig. 8b shows the outcome of grouping the buildings in the dataset into 17 clusters, the same number of clusters as available labels in the dataset. It notably segregates various residential buildings to a greater extent compared to the solution with 6 clusters, capturing the variation in the original labels more closely. However, it suggests a prioritization of certain similarities over those represented by the labels. If the objective of classification is to correctly categorize buildings according to the categories used in Norwegian building codes and standards, supervised learning appears to excel in assigning the correct labels compared to unsupervised k -means algorithms.

3.5. Testing of model on external datasets

To investigate the generalizability and transferability of the trained model, features and labels, the classification approach is tested on three

external datasets, published by researchers outside of the project, but relabelled by the authors for testing the classification methods developed in this paper. The purpose of this is to investigate whether the building categories/types and selected features are transferrable across countries and climate zones. The external datasets include the iFlex-dataset [39] with electricity time series data from residential buildings in Norway, the HUE-dataset [40] which is an hourly dataset of electricity use in Canadian residential buildings, and the BDG2-dataset [41] which includes non-residential buildings in the USA and the United Kingdom. For each external dataset the buildings were relabelled according to the labels presented in Table 1. The same 82 features were extracted from the electricity meter data for the external datasets. Different tests were conducted, where the original dataset presented in this article were used for training, and the external datasets were used for testing (and training).

3.5.1. Application on iFlex dataset (Norwegian households)

The iFlex dataset consists of a year of hourly AMS-measurements from over 2000 residential units (apartments and houses) in Norway from 2020 to 2021 [39]. The buildings were relabelled to the same labels used in Table 1 based on survey answers from the residents. The classification method was tested on the iFlex-dataset in two ways – first, using the original training set and the iFlex-dataset as the test set and

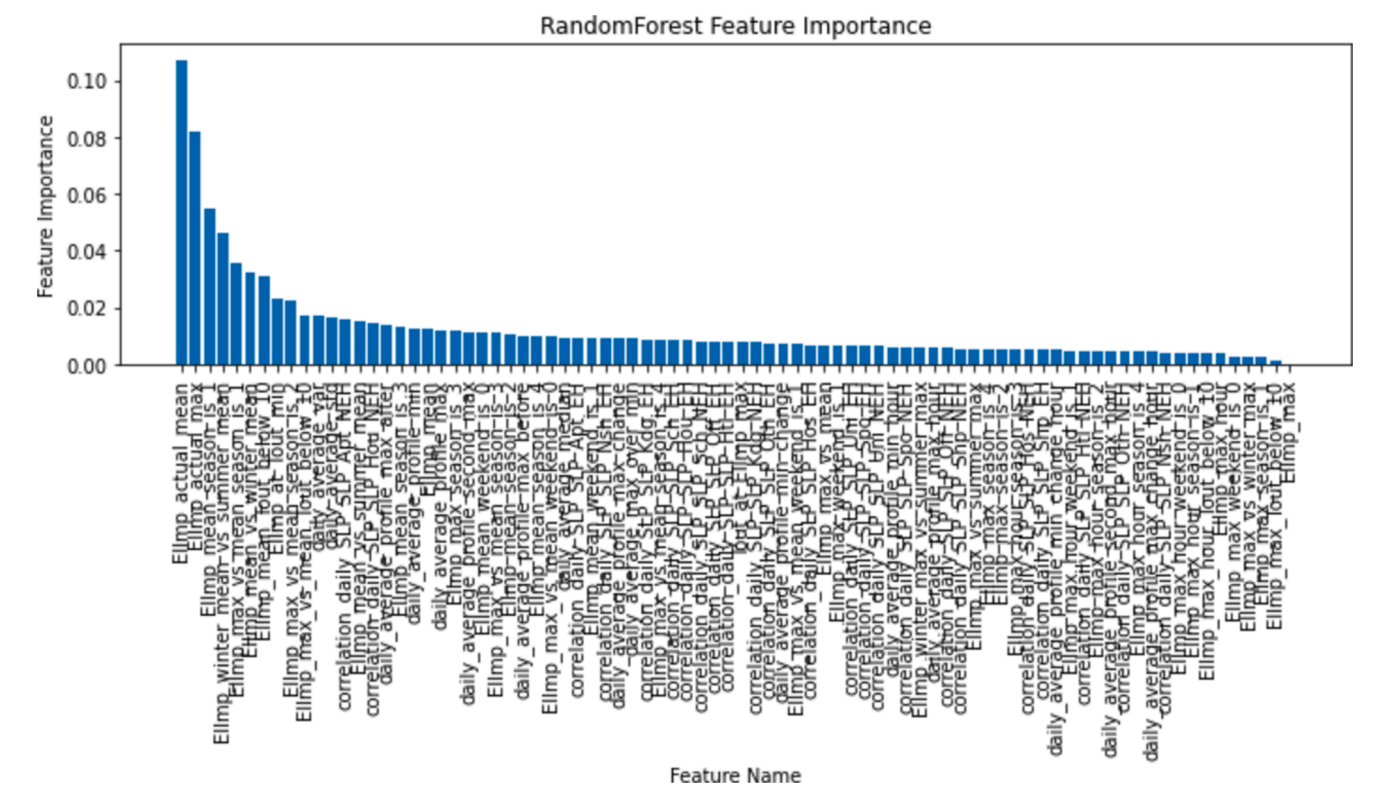


Fig. 4. Feature importance for Random Forest for building type classification.

Table 9
Performance of CatBoost and Random Forest on the test set when predicting only the building category.

Model	Accuracy	Accuracy Top 2	Precision	Recall	F1
Random Forest	0.892	0.967	0.895	0.892	0.891
CatBoost	0.888	0.98	0.894	0.888	0.888

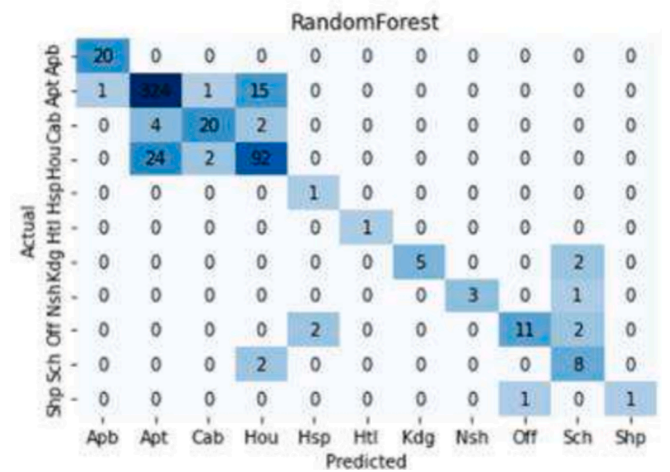


Fig. 5. Confusion matrix for Random Forest on the test set when predicting only the building category.

next, splitting the iFlex dataset into a training and testing set. Applying the method resulted in an accuracy of 0.7 when using the original dataset for the training and iFlex-dataset for the testing. When using the iFlex-dataset for both testing and training, the accuracy improved to 0.75. A reason for the performance not being higher for the iFlex-dataset

is that it only contains residential buildings, where the accuracy is lower compared to the accuracy for the entire dataset.

3.5.2. Application on HUE dataset (Canadian households)

HUE (The Hourly Usage of Energy Dataset for Buildings in British Columbia) is a dataset that comprises of hourly energy use measurements from 28 residential buildings in Canada (26 with sufficient labels). The buildings were relabelled as either “Hou” or “Apt”. The buildings in the HUE dataset were all correctly identified as residential buildings. The method however failed to correctly identify the correct building type in more than half of the cases and was unable to accurately separate between houses and apartments. When predicting only the building category (and not the heating type) all apartments are correctly predicted, while houses were mostly confused for being apartments as well. The top 2 accuracy was however perfect for predicting the building category.

3.5.3. Application on BDG2 dataset (USA and UK service buildings)

BDG2 (Buildings Data Genome 2) is a dataset with 2 years of hourly energy measurements (gas, cooling water, steam, and electricity) from 1636 non-residential buildings/rooms located in North America/Europe. Two tests were conducted with the BDG2-dataset. In the first test, a selection of buildings were relabelled based on their primary building category to the corresponding labels in Table 1. In the second test, the BDG2-dataset was used for both training and testing and without relabelling, but instead using the original “Primary use category” as labels. Some buildings were excluded due to missing data or labels.

4. Discussion

4.1. Results for classification of building type and category

This article has presented results for building type classification of electricity time series from buildings. RandomForest Classifier and the

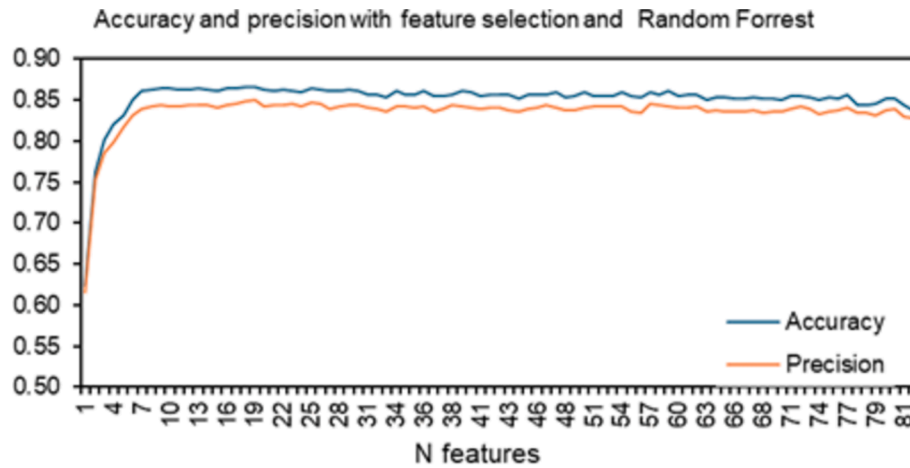


Fig. 6. Evolution of accuracy and precision for the building type classification using Random Forest Classifier and FSFS.

Table 10

Evolution of accuracy and precision for the building type classification using Random Forest Classifier and FSFS until the 9th feature is added.

N	Feature added	Accuracy	Precision
1	ElImp_actual_mean	0.624	0.616
2	+ daily_average_std	0.761	0.752
3	+ correlation_daily_SLP_SLP_Apt_EH	0.802	0.785
4	+ ElImp_max_vs_mean_weekend_is_1	0.820	0.799
5	+ ElImp_mean_vs_winter_mean	0.831	0.816
6	+ ElImp_actual_max	0.850	0.831
7	+ ElImp_max_weekend_is_1	0.861	0.839
8	+ ElImp_max_weekend_is_0	0.862	0.841
9	+ ElImp_max_season_is_1	0.864	0.843

CatBoost classifier give similar performance results, but the Random-Forest Classifier is a much faster approach and is hence preferred for this classification. The building type (building category and heating type) is correctly identified in 84 % of the cases, and with a 92 % top-2 accuracy. The accuracy reaches a peak of approximately 86 % when feature selection is used. This is a high and sufficient accuracy for this task, especially given as there is some uncertainty in the labels of the dataset, which is further elaborated in F.

Table 11

Performance of ensemble model, CatBoost and random forest when applied to the whole dataset and only residential buildings.

	Model	Acc.	Acc. top 2	Preci-sion	Recall	F1
All	Ensemble_model_soft (RF + CB)	0.844	0.917	0.831	0.844	0.833
	Ensemble_model_soft(RF + CB + GB)	0.840	0.919	0.827	0.840	0.830
Residential	RandomForest	0.861	0.935	0.836	0.861	0.848
	CatBoost	0.847	0.937	0.826	0.847	0.836
	Ensemble_model_soft (RF + CB)	0.858	0.937	0.833	0.858	0.846

Table 12

Performance of the method when applied to three external datasets (iFlex, HUE and BDG2).

		N	Target	Acc.	Acc. top 2	Precision	Recall	F1
iFLEX	Train: Original	2179	Building type	0.711	0.876	0.719	0.711	0.708
	Test: iFlex	1096						
	Train: 0.8 iFlex	876	Building type	0.750	0.923	0.743	0.750	0.745
	Test: 0.2 iFlex	220						
HUE	Train: Original	2179	Building category	0.462	1.000	0.821	0.462	0.439
	Test: HUE	26						
BDG2	Train: Original	2179	Building category	0.203	0.357	0.533	0.203	0.237
	Test: BDG2 (selected)	566						
	Train: 0.8 BDG2	1224	Primary categories from BDG2	0.565	0.755	0.519	0.565	0.521
	Test: 0.2 BDG2	306						

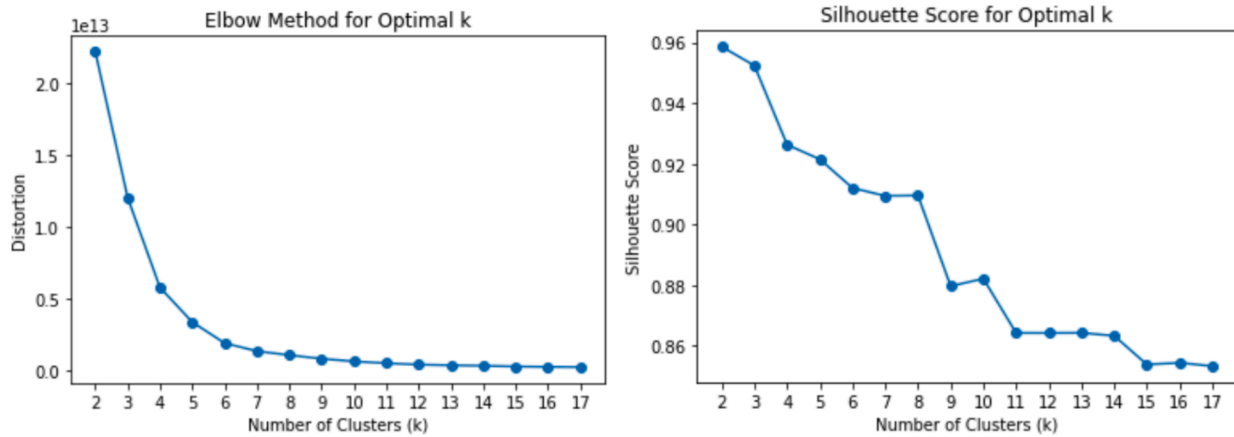


Fig. 7. (a) Distortion and (b) silhouette score for different numbers of clusters using k-means clustering.

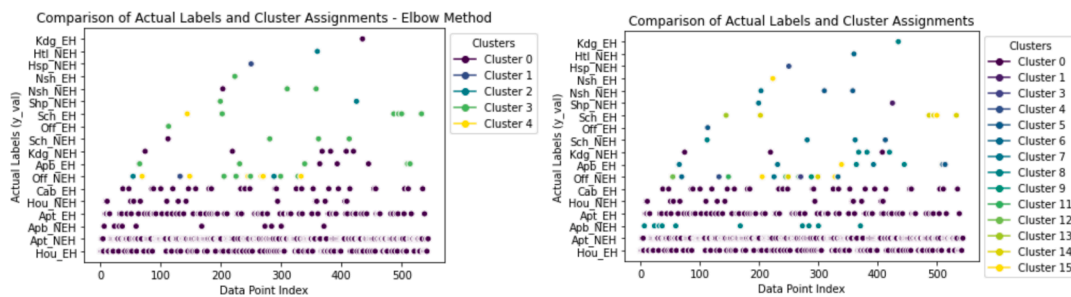


Fig. 8. Clustering of buildings in the dataset when using (a) 6 and (b) 17 clusters with k-means clustering. In both cases, the test set has no buildings assigned to one of the clusters.

article may lead to further improvement in accuracy. The performance will also vary slightly depending on which buildings are included in the training and test set.

4.3. Ensemble learning

While ensemble models are often powerful tools for improving accuracy, soft-voting ensemble models did not improve the performance compared to RandomForest and CatBoost classifiers alone in III. D. Ensemble models may not enhance performance due to several factors. These include redundancy and correlated errors among base models. This was to be expected due to the similarities in feature importances for these models but could not be ruled out without investigation.

4.4. Supervised vs. Unsupervised classification

The literature on building electricity load profiles shows that there are many examples of unsupervised clustering algorithms used for classification of buildings, efficiency classification of buildings and fault recognition of building electricity use. While unsupervised clustering of building load profiles are useful for many applications, including assigning pricing clusters of different costumers, supervised classification allows to get more tailored labels for the data, which can help efficiently collect metadata. In this article, buildings are labelled and classified according to the building categories used in the Norwegian building code. Achieving the same classification with unsupervised methods is not possible, as there may be similarities and differences between buildings of the predefined categories which will not automatically be captured by an algorithm without training on labelled data and the domain-specific features.

4.5. Testing on datasets across geography and climate zones

The classification method has exhibited good generalizability and transferability to Norwegian residential buildings, but poor transferability to the HUE and BDG2-datasets. There are a few possible explanations for this. First, the primary building categories used in the BDG2 dataset are broad categories with large disparities between the buildings within them. For example, the category “Education” contains the subcategories “School”, “Research”, “Student Centre”, “Classroom” among others, which are buildings which can be expected to have large disparities in user behaviour and energy systems, while the building categories in the original dataset contain more uniform buildings, making them easier to distinguish from each other. Secondly, the features extracted from the original dataset may not be the most suitable to predict the building category from other countries/climate zones and with different labels, for instance, looking at features more targeted at changes in the load caused by cooling and ventilation rather than heating may improve the results. Another reason may be that service buildings such as schools, may have user patterns that differ greatly in different countries for aspects such as needs for cooling, ventilation and heating, different control strategies and HVAC systems. Finally, there is a large uncertainty in the quality of the labels of the buildings in the BDG2-database, as [43] showed the way a building’s occupants use the spaces can be different than what was intended, which may cause misclassification or oversimplification. It was found that 26 % of the buildings in the original BDG-dataset was potentially mislabelled based on their load shape behaviour.

Onsite renewable energy production, such as photovoltaic (PV) panels mounted on buildings, can theoretically influence building load profiles by reducing the demand for imported electricity during periods of production. Although PV installations are becoming more common, most buildings in the database currently do not have PV systems. PV

generation typically peaks midday, especially during summer months. The key features identified by feature selection focus on both the average energy usage throughout the year and peak loads at different time periods. While mean and median electricity consumption might be affected by PV generation, peak loads, which generally occur during winter months, are less likely to be significantly impacted due to reduced solar radiation and lower PV output during the winter season. The extent to which local renewable generation influences a building's electricity import profile also depends on the ratio of energy production to consumption—the higher the production relative to consumption, the greater the impact on reducing imported electricity. While the classification of building type (building category and heating category) may not be largely affected by the presence of PV, disaggregation of electricity use for different appliances becomes more challenging without separate meters for PV generation, electricity export, and self-consumption, as total electricity consumption is necessary for the disaggregation task. Classifying buildings with and without PV is a task that could be considered for further research, if data from more producers are attained. The buildings' time series data include both smart meter data and climate data, including solar radiation. The method presented is adaptable, so given the availability of a dataset with meta data about PV-installations, the method could be adapted to classify the presence of PV-panels e.g. by adding features that consider the relationship between solar radiation and electricity consumption.

4.6. Data quality and labelling

The reliability of the labels in the dataset used in this article are not completely certain. The collection of metadata to be used for the labelling of the buildings has been conducted by several researchers using a variety of approaches, such as looking into building energy labels and the Norwegian building registry (cadastre), interviewing operational personnel, and building managers, and surveys.

The building categories of buildings in this database are the same as the building categories used in the Norwegian Standard NS 3031 "Energy and Power Demand for Heating of Buildings – Calculation Rules" [29] and for the energy requirements in the regulations on technical requirements for construction works within the Norwegian Building Acts and Regulations [16]. Some of these building categories contain buildings with large variations in energy consumption and user patterns, e.g. "Cultural buildings" which include buildings from churches to big concerts halls. The building categories used in these regulations differ from building types used in the cadastre, which divide buildings into more groups. Another issue with the labelling of buildings in the database is that buildings can only have one label, while a single building may have several uses, for example, a single building can consist of one part which is a school and another part which is an office but can only be assigned to one label in the database. Similarly, there is also some uncertainty in the labelling of buildings with "Electric Heating (EH)" or "Non-electric heating (NEH)". The metadata about the heating system of different buildings are acquired by different researchers with different approaches, and the while the metadata indicates the kind of heating system installed, it does not necessarily indicate if all the installed heating appliances are in use. For some buildings, there may also have been a change in the heating system during the duration of the measurements. One example is the use of oil and gas furnaces, which may have been in use until 2020 but exchanged later due to the fossil fuel ban for heating purposes in buildings that was enforced from January 2020. The use of heating appliances and electricity for heating may have been changed for several buildings with measurements that span across several years, but the label of heating appliances may not necessarily have been updated for all buildings in the database. The classifiers are trained only on certain building categories and cannot recognize other building categories not present in the dataset, such as, industrial buildings, parking houses, culture buildings, universities, sports buildings, and others. The model could however benefit from being able to

recognize if a building should not be assigned to any of the labels present and classify them as "Other" or "NA".

4.7. Consequences of misclassification

Predicting the wrong building category in the classification of electricity time series can have varying degrees of consequences depending on the subsequent use of the results. If utilized for labelling buildings before storing the data in a database for further reference, it is crucial to explicitly note that the assigned building category is an estimation. Moreover, it may be beneficial to provide the two most probable categories to mitigate the risk of propagating errors throughout subsequent processes such as energy pricing, integration into demand forecasting models, recommendations for energy-saving measures, and further analysis and research.

In the authors' ongoing research, the intention is to employ the predicted building category to develop a disaggregation algorithm for separating electric space heating loads from total building loads, with the predicted category serving as an input. If a building with electric heating is falsely classified as a building without electric heating using this method, the disaggregation algorithm may fail to accurately identify the electric heating energy usage as intended. Separating buildings with air-to-air heat pumps and buildings that use primarily electric panel heaters can be challenging on hourly data, as these are expected to exhibit similar patterns, especially during.

peak hours, but also have a slightly lower daily load profile on average. The model could be improved to estimate the heating type of buildings by adding more standard load profiles that differentiates between buildings with different building heating appliances.

4.8. Applications

As described in the introduction, the electrification of all end-use sectors is essential for transitioning to a sustainable low-carbon society. In Norway, the building sector consumes more than half of the nation's electricity consumptions, primarily for heating, with demand peaking during the coldest days of winter. With electrification, peak loads are expected to grow, and this raises the question on how we can limit the growth of peak loads. The suggested method has several applications and potential implementations that can aid in improving the knowledge on peak loads in buildings and areas, as well as on how to limit the peak load. By classifying the presence of electrical heating appliances, this approach offers a clearer understanding of energy consumption patterns across different building types. If the method is applied by energy providers, it could help identify and tailor energy efficiency and demand flexibility measures for electricity customers or analyse the use of different grid capacity tariffs. Additionally, the method could support improved grid and area planning by estimating the presence of specific customer types in a given region, which could help predict coincidence factors and peak loads. In data collection, the classification of building types provides metadata which can often be time-consuming to collect manually. The estimated class of a building could be used as an input feature in a disaggregation task, such as disaggregating electricity used for heating from a building's electricity load profile, thereby improving disaggregation accuracy. In the future, the method could be expanded to classify other forms of metadata, such as electric vehicle charging, photovoltaic systems, and heating types, further broadening its applications.

4.9. Future work

Future work includes enhancement and refinement of the presented methodology for classification of buildings. Firstly, one may benefit from expanding the current method to classify the type of electric heating systems present within buildings, which would provide valuable insights into energy usage patterns and aid in load disaggregation.

Similarly, the method could be extended to classify whether building is equipped with chargers for electric vehicles or not. This could be particularly useful for estimating the increasing prevalence of electric vehicles and their significant effect on building energy loads. Finally, to enhance the method's generalizability and applicability across other countries/climate zones, it could be beneficial to reevaluate the existing features, incorporate additional relevant features, and add additional data from various countries and climate zones to the training data set. Additional relevant features include pattern/shape based features, and possibly model based features [42]. Further, the method could be extended to predict the energy efficiency class of the buildings.

5. Conclusion

This article has introduced 82 domain-specific features and a supervised automatic building category classification approach for predicting the building category and heating type of Norwegian buildings from their electricity load profiles with high accuracy. While prior research has predominantly focused on unsupervised methods for classifying electricity consumers, the proposed approach has been tailored to label building load profiles according to Norwegian building regulations (TEK17) standard building categories, making it a more practical categorization which is useful for grid and area planning. The method was developed and tested on a large and original dataset of 2740 buildings across 10 building categories, including both residential and non-residential buildings, and achieved an accuracy of 84 % in identifying correct building and heating categories on the testing set. Furthermore, the proposed classification method demonstrates excellent generalizability and transferability when applied to unseen electricity data from Norwegian apartments. However, testing on buildings from other climate zones showed less transferability, indicating a need for more features tailored to different climate zones and locations.

CRedit authorship contribution statement

Synne Krekling Lien: Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Jayaprakash Rajasekharan:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Parts of the data is openly available, while other parts will be made available at a later time. Readers may request access to anonymized data from the authors.

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